

Hallucinators: A Preference-Concealing Negotiation Agent for the ANL 2026 League

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Abstract

We present *Hallucinators*, our agent for the ANAC 2026 Automated Negotiation League (ANL). The ANL 2026 score rewards two equally weighted objectives: securing a high-utility agreement (*Advantage*) and hiding one’s own preferences from the opponent (*Concealing*). *Hallucinators* addresses both through four complementary mechanisms: (i) a three-phase Boulware concession schedule, (ii) a distribution-based frequency opponent model with dynamic issue-weight refinement and behavioral profiling, (iii) an active preference-masking strategy, and (iv) extreme conflict detection with hardliner-advantage guards to manage adversarial negotiation domains. In a local Cartesian tournament against four standard baselines across all seven provided scenarios, *Hallucinators* achieved a mean final score of **1.372** (Advantage: 0.774, Concealing: 0.598), outperforming all competitors on both sub-components.

1 Introduction

The ANL 2026 competition evaluates bilateral negotiation agents on a *combined* score:

$$\text{Score} = \underbrace{(u(\omega) - r)}_{\text{Advantage}} + \underbrace{C}_{\text{Concealing}},$$

where $u(\omega)$ is the agent’s utility for the agreed outcome ω , r is its reservation value, and C is the fraction of the Kendall- τ ranking point not captured by the opponent’s frequency model of the agent. Maximising both objectives simultaneously is challenging: high-utility offers tend to be concentrated near the agent’s ideal point, which is precisely where the opponent’s model learns to look.

Our approach, *Hallucinators*, manages this tension by separating offer generation into a *utility-safe candidate set* and a *masking-optimal selection* within that set. Concession follows a three-phase Boulware schedule. A robust acceptance strategy with extreme-conflict rescue prevents sub-optimal agreements early on and deadlocks later, while a two-level candidate filter and RUFL-inspired phase policy select offers that balance own utility with opponent acceptance.

2 The Design of Hallucinators

Hallucinators extends the `BOANegotiator` base class from `NegMAS` and overrides the single `__call__` entry point, giving full control over acceptance and offering. All tunable constants are collected in a frozen `NegotiationConfig` dataclass (approx. 60 parameters), making the agent easily reproducible and tunable.

Offer management. At the start of each negotiation, `on_preferences_changed` enumerates all outcomes above the reservation value, sorts them by own utility, and caches them in `_outcomes`. The agent also precomputes a `_frontier_cache` of three named utility bands (`top_utility`, `safe_compromise`, `masked_rotation`) used by the late-game offer selection routines, avoiding repeated scans of the full outcome list during the negotiation. Additionally, all own utilities are stored in a sorted list to support fast percentile queries. The estimated opponent utility function is published in `private_info` (required by the ANL scoring infrastructure).

Three-phase concession. The target utility $T(t)$ at relative time $t \in [0, 1]$ follows:

$$\begin{aligned} t < 0.55 &\Rightarrow T = 0.92 u_{\max} \quad (\text{early, aggressive}), \\ 0.55 \leq t < 0.85 &\Rightarrow T = 0.92 u_{\max} - 0.17 u_{\max} \cdot \left(\frac{t-0.55}{0.30} \right)^{2.4} \quad (\text{mid, Boulware concession}), \\ t \geq 0.85 &\Rightarrow T = 0.75 u_{\max} - 0.11 u_{\max} \cdot \left(\frac{t-0.85}{0.15} \right)^{1.4} \quad (\text{late, final concession}), \end{aligned}$$

with a floor of $\max(r, 0.64 u_{\max})$. An opponent-responsive adjustment nudges the target up or down based on the observed concession rate and model confidence once 45% of time has elapsed. On the final step, the target utility is granted a strict relief of 2.5% $u_{\max}$ to secure last-second agreements.

3 Concealing Bidding Strategy

Our masking pipeline operates in three layers.

Issue-cost estimation. At initialisation we estimate each issue’s *cost* — how much own utility varies across its values. Issues with low cost are “cheap” (we can vary them freely); issues with high cost are “expensive” (fixing them near our optimum is essential).

Decoy signalling. During the early and mid phases ($t < 0.78$) each candidate offer is scored by a *decoy signal*:

$$s_{\text{decoy}} = 0.65 \cdot \text{cheap_stability} + 0.35 \cdot \text{expensive_variety},$$

where *cheap_stability* rewards keeping cheap-issue values the same as the previous offer (signalling *fake* resistance), and *expensive_variety* rewards changing expensive-issue values (hiding the high-cost dimension).

Masked window selection. After scoring all candidates, we form a *masked window*: the subset of top candidates whose own utility falls within a narrow slack band (3.5% early, 7% late) around the best candidate. Within this window, offers are re-ranked by masking score — issue novelty, exact-repeat avoidance, and recent-value reuse — to maximally confuse the opponent’s frequency model.

Two-level candidate filtering and RUFL policies. Before multi-objective scoring, the candidate pool undergoes a two-level selection pass active from the mid game ($t \geq 0.60$). First, we preserve a utility-safe band (within 3.5% of the best available utility), then we rank these safe candidates by their estimated acceptance likelihood, keeping the top 60%. Additionally, RUFL-inspired phase-specific policies are applied late in the negotiation. Near the absolute deadline ($t \geq 0.94$), the regular selection is overridden entirely to explicitly propose the best offer previously made by the opponent, or bids with extreme opponent-acceptance likelihoods, bypassing masking completely to avert timeouts.

4 Acceptance Strategy

Acceptance is checked in the following priority order.

1. **Reservation guard.** Reject any offer below r .

2. **Extreme conflict rescue.** If the domain is tiny and the negotiation is stalled, we attempt an extreme-conflict rescue. We accept the opponent’s best offer seen so far if it is within 2% of our own rescue compromise. However, if the opponent is identified as a strong hardliner, this rescue is bypassed to prevent exploitation.
3. **ACcombi** (Baarslag et al., 2010). At $t \geq 0.99$, accept if the offer utility is at least the window-average of recent opponent offers.
4. **Target utility.** Accept if $u(\text{offer}) \geq T(t)$.
5. **Last-chance.** On the final step, accept the current offer if it is at least as good as the best offer the opponent has made so far and meets $0.98 T(t)$.
6. **Counter-offer margin.** Accept if $u(\text{offer}) \geq (0.995 - 0.08 t) \cdot u(\text{planned})$.
7. **Late-game floor.** At $t > 0.97$ accept any offer above the emergency floor $0.55 u_{\max}$ (raised to $0.62 u_{\max}$ if the hardliner advantage guard is active).

5 Opponent Model

Distribution-based frequency model. The `DistributionAnalyzer` maintains per-issue weighted frequency counts $C_i(v)$ of opponent offers. The estimated utility of an outcome ω is a weighted sum of per-issue value scores:

$$\hat{u}_{\text{opp}}(\omega) = \sum_i w_i \cdot \left(\frac{C_i(\omega_i) + \lambda}{C_i^* + \lambda} \right)^{0.35},$$

where $\lambda=1$ is a Laplacian smoothing factor, C_i^* is the maximum count over values of issue i , and w_i is the estimated issue weight. The power exponent 0.35 compresses high-frequency signals to reduce overconfidence. Earlier offers have higher model weight; repeated identical offers are down-weighted exponentially (damping factor 0.48, minimum weight 0.18).

Issue-weight update. Every $k=4$ offers we compare the L_1 distance between the previous and current window distributions. Issues that remain *stable* in a window where the opponent conceded on at least one other issue are inferred to be high-priority; their weights are boosted by $0.08(1 - t)$ and the vector is renormalised.

Behavioral profiling and Acceptance likelihood. Beyond frequency modeling, the agent maintains an active four-component behavioral profile of the opponent, estimating whether they act as a *hardliner*, a *conceder*, a *fixed-threshold acceptor*, or a *mirror strategist*. These profiling scores modulate concession aggressiveness and extreme-conflict rescue guards. Furthermore, they directly inform our estimated acceptance likelihood (s_{acc}) for candidate offers. The acceptance likelihood integrates the frequency-based opponent score with the offer’s structural similarity to past opponent bids, while applying dynamic bonuses for hardliners (who demand high similarity) and patience-based adjustments as the deadline approaches.

Opponent deadline forecasting and Lookahead. To evaluate candidate offers under uncertainty, we predict the best utility the opponent may offer by the deadline, adapting a utility-fit approach from Team 271. We compute a median over pairwise concession slopes to establish a robust trend line, and scale it by the remaining time. This forecast drives a one-step lookahead tiebreak score ($s_{\text{lookahead}}$) and determines our confidence in the opponent model.

Multi-objective candidate scoring. Each candidate offer is scored as:

$$s = w_{\text{own}} s_{\text{own}} + w_{\text{acc}} s_{\text{acc}} + w_{\text{par}} s_{\text{par}} + w_{\text{mask}} s_{\text{mask}} + 0.04 s_{\text{deadline}} + 0.035 s_{\text{decoy}} + 0.025 s_{\text{lookahead}},$$

with phase-specific weight vectors ($w_{\text{own}}, w_{\text{acc}}, w_{\text{par}}, w_{\text{mask}}$): early (0.74, 0.06, 0.04, 0.16), mid (0.60, 0.16, 0.10, 0.14), late (0.48, 0.28, 0.16, 0.08).

6 Evaluation

We evaluated Hallucinators in a Cartesian round-robin tournament against four standard NegMAS baselines across all seven provided scenarios (Amsterdam, Camera, Car, Grocery, ISBTAcquisition, Laptop, NiceOrDie), yielding 140 negotiations per agent.

Table 1: Mean tournament scores (Advantage + Concealing = Score). Higher is better.

Agent	Advantage	Concealing	Score
Hallucinators (ours)	0.774	0.598	1.372
BOANeg	0.580	0.594	1.174
BoulwareTBNegotiator	0.578	0.000	0.578

Advantage. Hallucinators achieves a mean Advantage of **0.774**, compared to 0.580 for BOANeg. The three-phase Boulware schedule combined with the robust target-utility tracking prevents the agent from accepting sub-par offers early in the negotiation, preserving surplus for later. The two-level candidate filter and RUFL phase policy further improve the selected bids, consistently extracting high utility while maintaining high acceptance probabilities.

Concealing. The Concealing score of **0.598** reflects the effectiveness of the decoy-signal and masked-window strategies in confusing the opponent’s frequency model. BoulwareTBNegotiator scores 0.000 because it does not publish an `opponent_ufun` in `private_info`, and the tournament infrastructure therefore cannot score its opponent model.

7 Lessons and Suggestions

1. **Guarding against hardliners pays off significantly.** The hardliner advantage guard was a crucial contributor to improved Advantage. Without it, the extreme-conflict rescue mechanism frequently conceded too much to opponents who simply refused to move. Bypassing the rescue in these cases forced more equitable outcomes.
2. **Two-level candidate selection balances utility and acceptance.** Filtering the top candidates by a strict own-utility slack (3.5%) and then keeping the top 60% ranked by opponent acceptance likelihood ensured that the agent only offered highly acceptable bids without sacrificing its own Advantage.
3. **Precomputed frontier bands save significant computation.** In large outcome spaces the late-game routines repeatedly iterated over thousands of outcomes. Caching three utility bands at initialisation reduced negotiation latency and avoided worst-case slowdowns in deadline-range scenarios.
4. **Require strong evidence before acting on opponent models.** The responsive concession and median defense both require a minimum confidence level. Premature activation on noisy early data caused erratic mid-game behaviour in earlier agent versions.

Conclusions

Hallucinators demonstrates that the ANL 2026 dual objective of maximising agreement utility while concealing preferences can be addressed through a unified pipeline. By separating candidate generation (utility-safe) from candidate selection (masking-optimal), adding a two-level filter to prioritize acceptance, and utilizing behavioral profiling alongside hardliner-advantage guards for extreme conflicts, the agent achieves strong performance on both the Advantage and Concealing components. Future work will explore learned models of opponent types and multi-round evidence aggregation.

References

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